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Editors
Transportation Research Part E: Logistics and Transportation Review

Dear Editors,

Please consider the enclosed manuscript, "Cost, Carbon, and the Speed of Replenishment in Distribution Network Design," for publication in *Transportation Research Part E: Logistics and Transportation Review*.

Firms are redesigning their distribution networks on two fronts, moving stock closer to demand and slowing their line haul. The reasons they give are mainly cost and delivery service; carbon pricing and emissions reporting now bear on such decisions as well. The manuscript models the two decisions jointly. A firm chooses how many distribution centers to operate and how fast to replenish them, under a carbon price or cap and a service requirement. Slower line haul burns less fuel but requires more safety stock; more distribution centers shorten the last mile but add facilities and give up the pooling that keeps inventory low.

We show that the design problem has a single best solution at every carbon price, and that the carbon-minimizing direction of both decisions follows from four carbon-intensity-per-dollar ratios and the share of a distribution center's cost that is lost pooling. Because faster replenishment recovers part of the pooling that dispersion gives up, the carbon intensity of line-haul freight bears on how many distribution centers to operate. Calibrated to public US emission factors and logistics costs, the direction depends on how speed is bought and how the last mile is driven. Speed bought with fuel is slowed under carbon pricing, speed bought with labor is favored, and heavy diesel delivery calls for more distribution centers while electric vans call for fewer. The same ratios show that a carbon instrument covering transport alone can raise total logistics carbon.

The manuscript fits the journal's focus on logistics and operations models with application and on the evaluation of public policies. It is original, has not been published, and is not under consideration elsewhere.

Sincerely,

Ram Ganeshan